

JOSHUA WOLFF ANDERSON

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SUMMARY

PhD Candidate in Intelligent Systems with a strong foundation in data science, specializing in bias, interpretability, and causal methods to develop fair, transparent, and effective models for real-world challenges in healthcare and international development. Current research focuses on bias mitigation in clinical decision-making. Passionate about leveraging AI to improve quality of life through robust, transparent, and practical implementations.

EDUCATION

PhD	University of Pittsburgh, Intelligent Systems	Expected: Spring 2027
MS	Chapman University, Computational and Data Sciences	May 2022
BS	Chapman University, Computer Science	May 2020

TEACHING EXPERIENCE

Courses

- Intro to Python, Hillman Academy/CoSBBi Summer 2026
- Intro to Artificial Intelligence, Hillman Academy/CoSBBi Summer 2026

Guest Lectures

- Machine Learning for Managers, Chapman University Fall 2025
 - Topics in Machine Learning: Algorithmic Fairness
- High Performance Computing, Chapman University Fall 2025
 - Topics in Artificial Intelligence: Mechanistic Interpretability
- Machine Learning for Managers, Chapman University Spring 2025
 - Topics in Machine Learning: Algorithmic Fairness
- Intro to Data Science, Chapman University Spring 2022
 - Topics in Data Science: Causal Inference

Teaching Assistant

- Data Structures and Algorithms, Chapman University Spring 2022
- Data Structures and Algorithms, Chapman University Fall 2021
- Machine Learning for Managers, Chapman University Fall 2020
- Statistical Models for Business, Chapman University Fall 2020

Trainings

- Center for Integration of Research, Teaching and Learning (CIRTL) Spring 2026
Associate Certification in STEM teaching

WORK EXPERIENCE

NIH T15 Research Fellow Jan. 2023 to Present

University of Pittsburgh, Pittsburgh, PA

- Conduct research and present on bias in clinical predictive modeling to both researchers and non-technical clinicians
- Developed a decision-theoretic framework that revealed mitigating bias on probability-based metrics can improve prediction scores without meaningfully improving utility of patient outcomes
- Published multiple papers for the Center for Clinical Artificial Intelligence (CCAI)

Data Scientist & Graduate Research Assistant

Aug. 2020 to Aug. 2022

World Bank & Chapman University, Orange, CA

- Built a population vulnerability index for sub-Saharan Africa fusing five heterogeneous geospatial data sources (e.g., violent conflict, natural disaster risk, proximity to public services)
- Modeled a single actionable score to guide World Bank development-funding decisions
- Engineered geospatial features by aggregating raster and point data (climate, conflict, facility access) to common administrative boundaries for index construction

Applications Development & Vulnerability Analyst

Mar. 2019 to Aug. 2019

CISOSHARE, San Juan Capistrano, CA

- Built and debugged web application modules using React.js, interfacing with RethinkDB
- Applied cybersecurity concepts, such as application security layers and logging

Data Analyst

Jun. 2018 to Aug. 2018

Ingram Micro, Irvine, CA

- Built an automated inventory-audit pipeline using SQL and a server-based application to reconcile specific inventory counts and flag anomalies for review, replacing a manual process

Computer Science & Mathematics Tutor

Feb. 2017 to Jul. 2017

Oxford Tutoring, Irvine, CA

- Tutored AP Computer Science students to prepare them for the AP exam
- Taught mathematics to grade school students to accelerate their education
- Debugged and mentored student projects to help them learn Java

ACADEMIC SERVICE

Peer Reviewer

- ACM Transactions on Intelligent Systems and Technology
- NeurIPS Reliable ML Workshop
- International Conference on Information Systems
- Journal of American Medical Informatics Association
- Journal of Biomedical Informatics

PUBLICATIONS

Current Projects

- Hybrid novice–AI system achieves expert-level performance in intraoperative ischemia detection (In review, npj Digital Health)
- A Decision-Theoretic Perspective on Fairness in Clinical Predictive Models (In review, npj Health Systems)
- A Framework for Data-Agnostic Counterfactual Fairness in Transformers through Activation Steering (In preparation)

Awards

- AMIA Informatics Summit 2026, *Student Paper Award, 2nd Place*: “The Effect of Enforcing Fairness on Reshaping Explanations in Machine Learning Models.”
- AMIA Informatics Summit 2024, *Student Paper Award finalist*: “Measuring and Reducing Racial Bias in a Pediatric Urinary Tract Infection Model”
- AMIA Annual Symposium 2023, *Distinguished Poster Award finalist*: “Assessing Racial Bias in Clinical Prediction for Urinary Tract Infections”

Papers

- Anderson J. W., Visweswaran S. (2026) The Effect of Enforcing Fairness on Reshaping Explanations in Machine Learning Models. *AMIA Summits on Translational Science Proceedings*, 2026, 47.
- Anderson, J. W., & Visweswaran, S. (2025). Algorithmic individual fairness and healthcare: a scoping review. *JAMIA Open*, 8(1), ooae149.
- Anderson, J. W., Shaikh, N., & Visweswaran, S. (2024). Measuring and reducing racial bias in a pediatric urinary tract infection model. *AMIA Summits on Translational Science Proceedings*, 2024, 488.
- Anderson, J. W., & Rakovsk, C. (2023). An R package for parametric estimation of causal effects. *arXiv Preprint arXiv:2307.08686*.
- Anderson, J. W. (2022). CausalModels: An R Library for Estimating Causal Effects. *Chapman University*.

- Anderson, J. W., Encina, L. I. A., Karippacheril, T. G., Hersh, J., & Stringer, C. (2022). A Topic Modeling Approach to Classifying Open Street Map Health Clinics and Schools in Sub-Saharan Africa. *arXiv Preprint arXiv:2212.12084*.

Posters & Presentations

- Anderson, J. W. The Effect of Enforcing Fairness on Reshaping Explanations in Machine Learning Models. *UPMC Digital Health Summit*. 2026 Jan 29.
- Samayamuthu, M. J., Sadhu, E., Anderson, J. W., Visweswaran, S. A Survey of Clinical Algorithms with Race. *AMIA Annual Symposium Proceedings*. 2024 Mar. 20.
- Anderson J.W., Shaikh N., Visweswaran S. Assessing racial bias in clinical prediction for urinary tract infections. *AMIA Annual Symposium Proceedings*. 2023 Nov 13. (*Finalist for the Best Poster Award at the AMIA Annual Symposium, 2023*)
- Anderson J.W., Shaikh N., Visweswaran S. Measuring and mitigating racial bias in predicting pediatric urinary tract infections. *Proc. IEEE-EMBS Int. Conf. Biomed. Health Informat. (BHI)* 2023 Oct 15.
- Anderson, J. W., Shaikh, N., Visweswaran, S. Reducing Racial Bias in Clinical Models for Predicting Urinary Tract Infections. *UPMC Digital Health Summit*. 2023 Oct 4.